THE SHEAF WITH EXACT SUPPORT OF A DIVISOR IS THE TENSOR PRODUCT OF THE PULLBACK WITH THE SHIFTED CONORMAL SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AA4 (sources/stacks-duality/pinned-original-source.tex line 3445).

Allowed standard prerequisites (editorial).

- The computation of the internal Hom out of the structure sheaf of a divisor. Same chapter, Lemma labelled lemma-compute-for-effective-Cartier, cited at proof lines 3446-3448 and 3464-3468. Supplies the identification with the shifted conormal sheaf.
- The projection formula for a closed immersion used to define the map. Cohomology, Lemma labelled cohomology-lemma-projection-formula-closed-immersion, cited at proof lines 3470-3474. Rewrites the left hand side after pushforward.
- Duality for perfect complexes. Cohomology, Lemma labelled cohomology-lemma-dual-perfect-complex, cited at proof lines 3480-3486. Gives the final equality on the right hand side.

Source reference. Original equation 0B4C: sources/stacks-duality/pinned-original-source.tex, source lines 3411–3411. Same-chapter pointer into the author environment equation at source line 3411; invoked at proof line 3468 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B4B: sources/stacks-duality/pinned-original-source.tex, source lines 3368–3368. Same-chapter pointer into the author environment lemma at source line 3368; invoked at proof line 3465 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0B55: sources/stacks-chow/tags.txt, source lines 14023–14023. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3467 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3483 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *As above, let X be a scheme and let $D \subset X$ be an effective Cartier divisor. Then (0B4C) combined with Lemma 0B4B defines an isomorphism*

$$Li^*K \otimes^{\mathbf{L}}_{\mathcal{O}_D} \mathcal{N}[-1] \longrightarrow R\mathcal{H}om(\mathcal{O}_D, K)$$

functorial in K in $D(\mathcal{O}_X)$.

Proof. Since i_* is exact and fully faithful on modules, to prove the map is an isomorphism, it suffices to show that it is an isomorphism after applying i_*. We will use the short exact sequences $0 \to \mathcal{I} \to \mathcal{O}_X \to i_*\mathcal{O}_D \to 0$ and $0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to i_*\mathcal{N} \to 0$ used in the proof of Lemma 0B4B without further mention. By Cohomology, Lemma 0B55 which was used to define the map (0B4C) the left hand side becomes

$$K \otimes^{\mathbf{L}}_{\mathcal{O}_X} i_*\mathcal{N}[-1] = K \otimes^{\mathbf{L}}_{\mathcal{O}_X} (\mathcal{O}_X \to \mathcal{O}_X(D))$$

The right hand side becomes

$$\begin{aligned} R\mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_D, K) &= R\mathcal{H}om_{\mathcal{O}_X}((\mathcal{I} \to \mathcal{O}_X), K) \\ &= R\mathcal{H}om_{\mathcal{O}_X}((\mathcal{I} \to \mathcal{O}_X), \mathcal{O}_X) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K \end{aligned}$$

the final equality by Cohomology, Lemma 08DQ. Since the map comes from the isomorphism

$$R\mathcal{H}om_{\mathcal{O}_X}((\mathcal{I} \to \mathcal{O}_X), \mathcal{O}_X) = (\mathcal{O}_X \to \mathcal{O}_X(D))$$

the lemma is clear. □